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I build financial software. Shipped, in production, with the tax math unit-tested.

Shipped

NOSTRO: BATCH RECONCILIATION OF FOREIGN REMITTANCES, WITH A MEASURED ACCURACY HARNESS

Solo · TypeScript · Aug to Sep 2026 · Razorpay AI Buildathon, Track 04

Every foreign bank credit an Indian exporter receives must be tied to an invoice *and* a FIRA before the revenue counts as a zero-rated export. Nostro runs that reconciliation as a batch job and reports what it could not resolve as carefully as what it could.

- **250 invoices, twelve noise classes** (split and merged remittances, FX drift across a rate month, flat SWIFT fees, missing certificates, wrong purpose codes). **96.52% allocation precision, 81.62% recall, ~51,000 records/sec.** One command reproduces every number from a seed.
- **₹0.00 under-declared.** The rupee value of anything wrongly called zero-rated. A run that under-declares exits non-zero regardless of its F1. That metric, not accuracy, is the design goal.
- **Ambiguity is refused, never guessed.** 151 exceptions across 7 reason codes, each carrying the exposure it represents, so the queue sorts by what it costs to be wrong about.
- **The layer boundary is enforced by a test:** `src/core/` has no network dependency and cannot import the model layer. The LLM sees only narrations the deterministic parser flags for review, returns a schema-validated hint, and never makes an allocation.
- **110 unit tests**, strict typecheck, none requiring an API key.

RAHA: GST AND TAX COMPLIANCE FOR INDIAN CREATORS EARNING FOREIGN INCOME

Solo founder and sole engineer · live at raha.software · Feb to Aug 2026

Indian freelancers and creators who invoice foreign clients must prove each payment is an *export of service* to claim zero-rated GST. Miss the proof and the revenue is reclassified as domestic, and that is 18% GST on money already spent. Raha closes that loop.

- **9,200 lines of TypeScript**, 67 commits, in production on Vercel + Supabase (Mumbai region, for financial data localisation).
- **Every rupee-affecting calculation is a pure, unit-tested function with no I/O.** RBI reference-rate FX conversion, FIRA↔income reconciliation, GST invoice math, advance-tax and GST-liability estimation, export-status classification. Money math does not get to be probabilistic.
- **FIRA reconciliation engine:** matches foreign bank credits to income entries on a $-7/+45$ day window with $\pm 2\%$ amount tolerance (bank charges and forex spreads move the INR figure), handles partial and split allocations, and refuses to assert a match the user has not confirmed.
- **Row-level security on every table**, storage policies for the document vault, Zod validation at every boundary, service-role keys server-side only.
- Next.js 15 (App Router) · TypeScript · Supabase (Postgres + RLS + storage) · Zod · Vitest · Playwright · Tailwind + shadcn/ui.
- Wrote the domain research myself, covering LUT gating, purpose codes, Sec 2(6) IGST export treatment and the 2026 creator profession code, sourced from primary tax guidance rather than vibes.

NEXUS: PROVENANCE-AWARE PERSISTENT MEMORY FOR AI SYSTEMS

Solo · research project · Python · 2026

Testing whether a structured, provenance-tracked world model beats conversation history and naive vector retrieval at long-horizon task continuation. Memory records, relationship graph, repositories and lexical retrieval implemented and tested; `mypy --strict` **clean**. Unbuilt subsystems are reserved as empty packages rather than shipped half-done, and the README status table says exactly what is and is not implemented.

How I work with AI

I use AI as an engineering team, not autocomplete. Spec, plan, implement, review, ship, with skill libraries and agent harnesses driving the loop. Both projects above were built this way.

The judgment that matters is knowing where **not** to put a model. In Raha, no LLM touches a number that becomes a tax liability. Language models parse messy bank narrations and extract PDF fields; deterministic, tested functions decide what is owed.

Education

Indian Institute of Technology Madras. B.Tech, Chemical Engineering · expected Aug 2028
Relevant coursework: Computational Programming & Process Simulation, Computational Techniques, Statistical Design and Analysis of Experiments, Innovation & Entrepreneurship.

Stack

TypeScript · Python · Next.js · React · Postgres · Supabase · Zod · Vitest · Playwright · Tailwind
· Git · Vercel · LLM tooling and agent harnesses